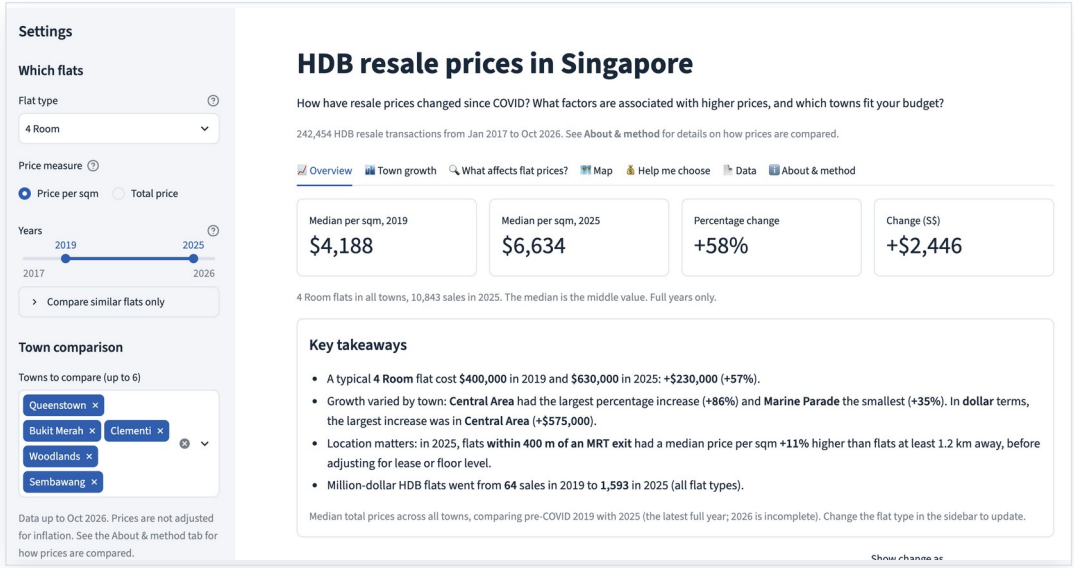
DS608 DATA ANALYTICS FOR ECONOMICS · GROUP PROJECT

Where did HDB resale prices rise most after COVID?

An interactive dashboard of Singapore resale flat transactions, 2017–2026



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Singapore Management University, MSc in Data Science · 10 November 2026

MOTIVATION AND QUESTIONS

Three questions about HDB resale prices

As international students, the first thing we had to do in Singapore was find a place to rent. Rent depends on how expensive an area is, and we kept hearing that prices jumped after COVID.

Data

HDB resale transactions, Jan 2017 to Oct 2026, about 242,000 sales (data.gov.sg)

MRT exit locations (LTA) and block locations (NUS 3D HDB dataset, OneMap)

1 **Growth**

Which towns' prices rose most after COVID (2019 compared with later years)?

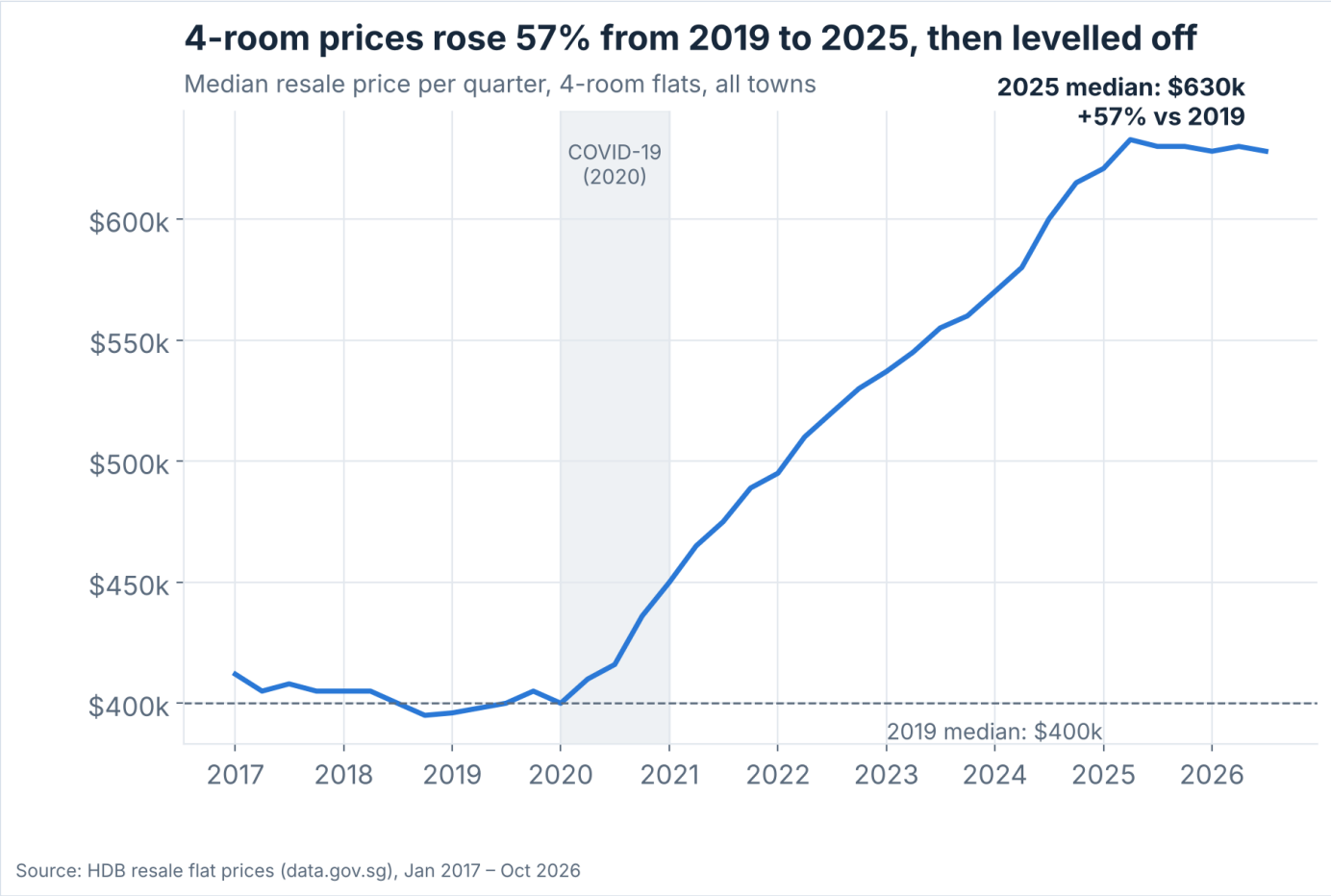
2 **Drivers**

Why are some flats and locations worth more: lease, floor, size, distance to the MRT and the city centre?

3 **Decision support**

Can a dashboard help someone find towns and blocks that fit their budget?

1 · PRICES AFTER COVID



Source: HDB resale flat prices (data.gov.sg), Jan 2017 – Oct 2026

+57%

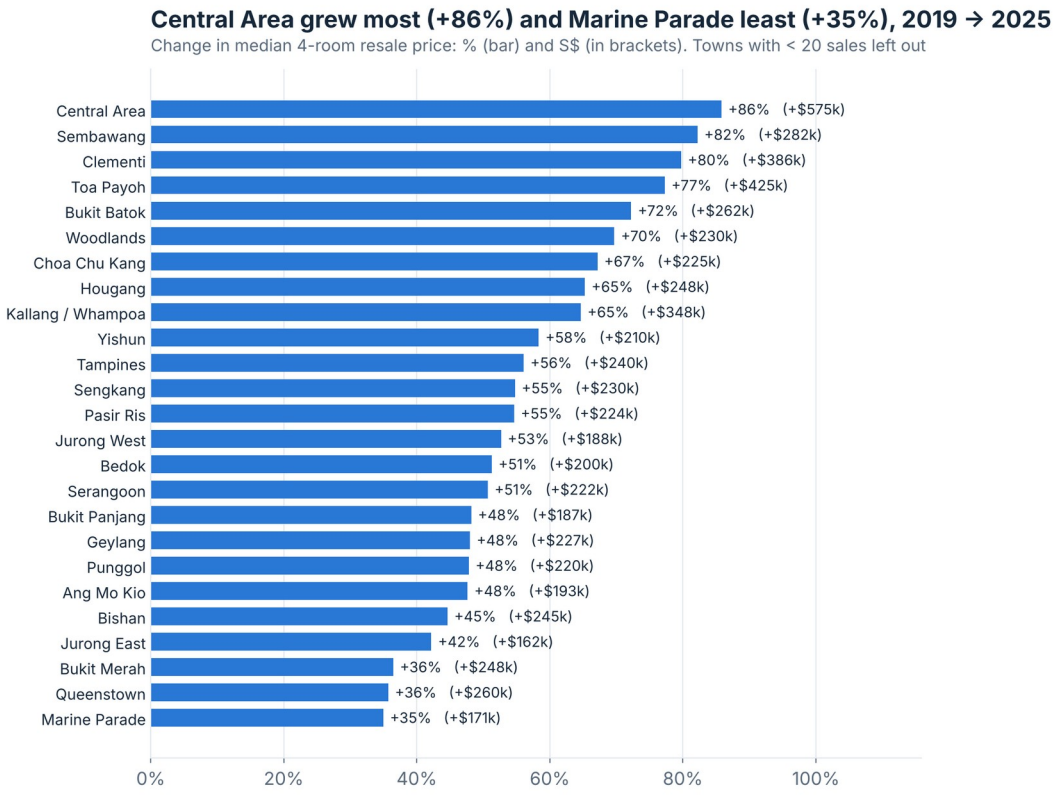
Median 4-room price, 2019 to 2025: $400k to $630k

Price per sqm rose by about the same: $4,188 to $6,634 (+58%).

Prices hardly moved before 2020, rose fast until mid-2025, and have stayed near $630k since.

2026 covers Jan to Oct only, so year-to-year comparisons stop at 2025.

2 · WHICH TOWNS GREW MOST

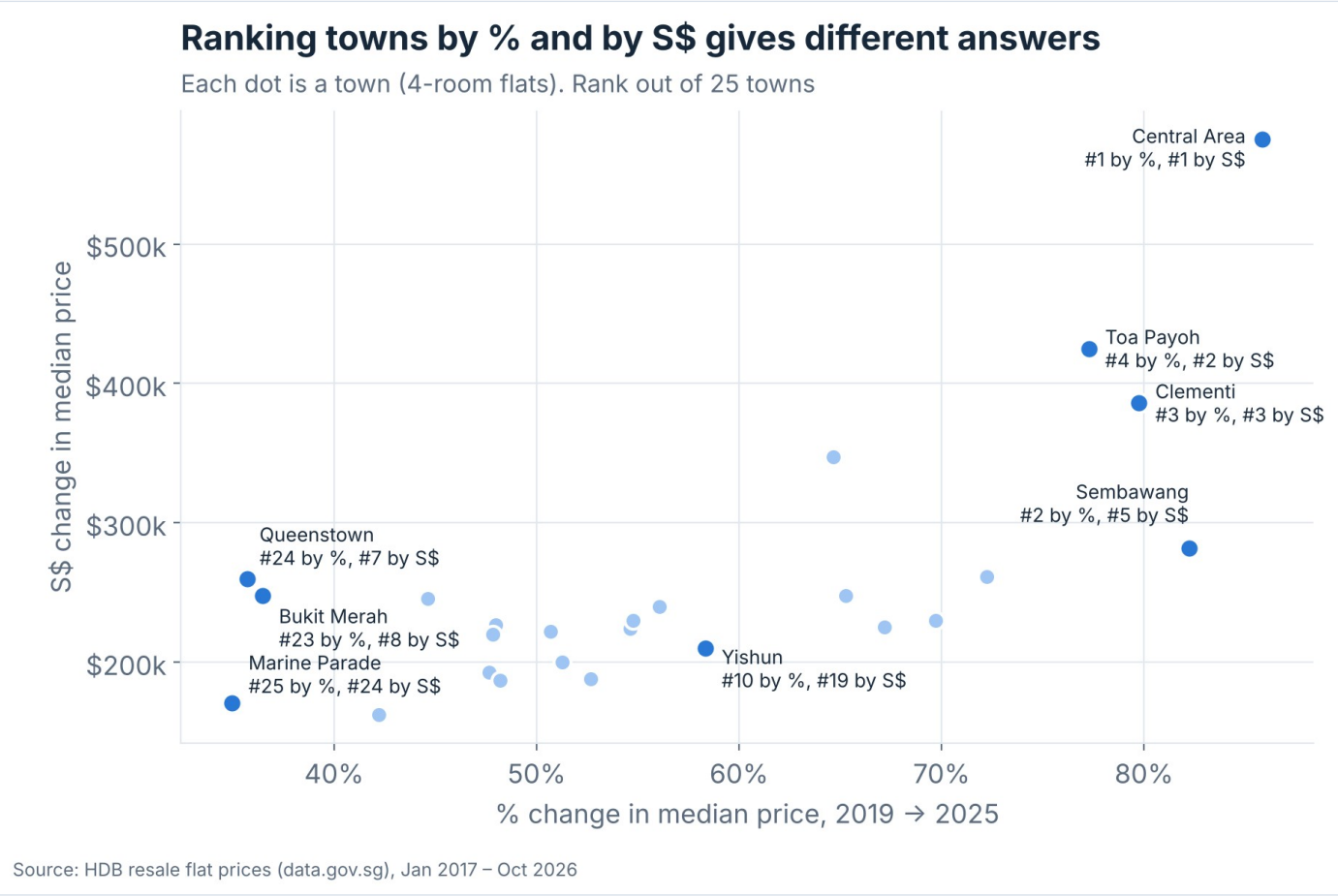


Fastest growth	% change	S$ change
Central Area	+86%	+$575k
Sembawang	+82%	+$282k
Clementi	+80%	+$386k
Toa Payoh	+77%	+$425k

Slowest growth	% change	S$ change
Marine Parade	+35%	+$171k
Queenstown	+36%	+$260k
Bukit Merah	+36%	+$248k

Growth was uneven: Central Area’s +86% is about 2.5 times Marine Parade’s +35%.

4-room flats; towns with fewer than 20 sales in either year are left out (25 towns remain).



24th vs 7th

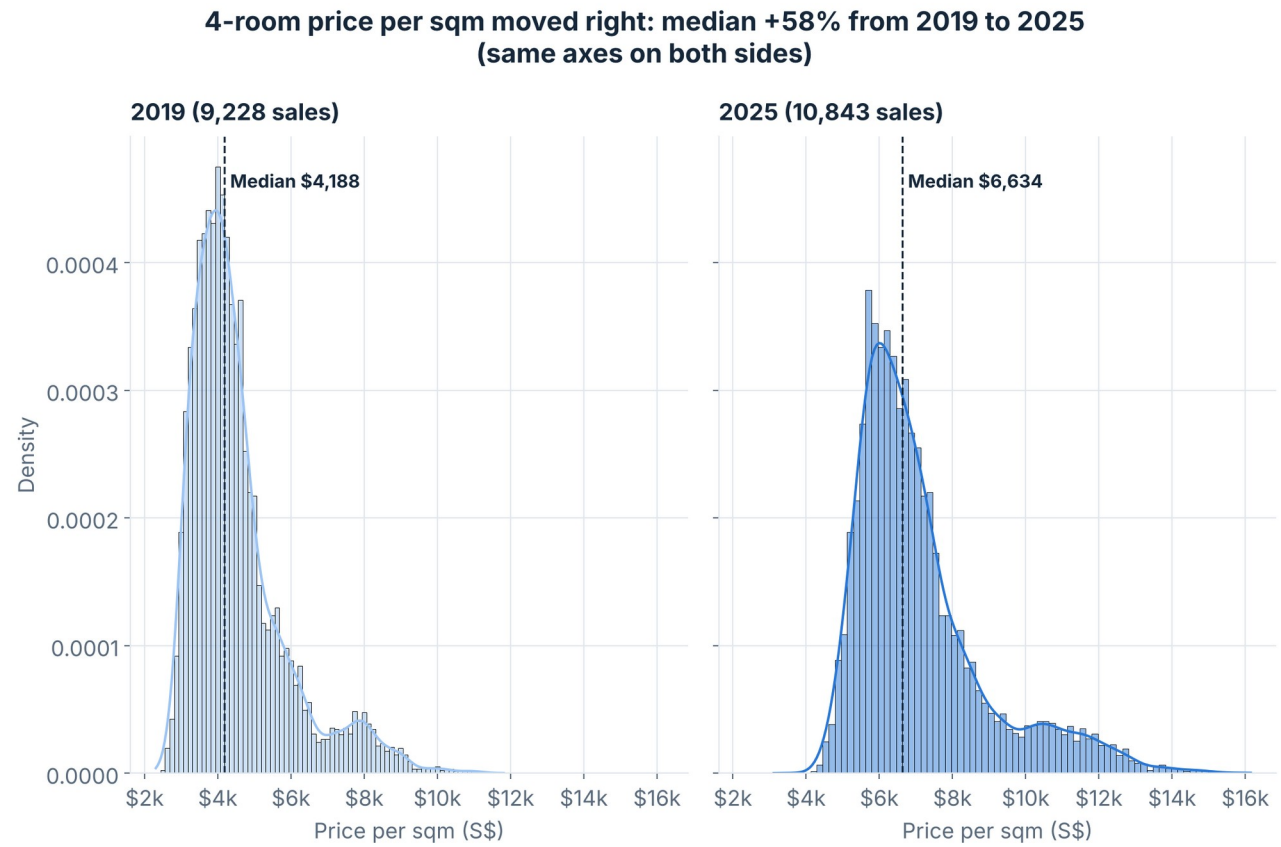
Queenstown's rank by % change and by S$ change, out of 25 towns

Queenstown started high ($728k) and added $260k: only +36%.

Sembawang started low ($343k) and added $282k: +82%, 2nd by % but 5th by S$.

For a buyer, the dollar change is the extra money to pay or borrow. The dashboard shows both.

4 · THE WHOLE RANGE MOVED UP



64 to 1,593

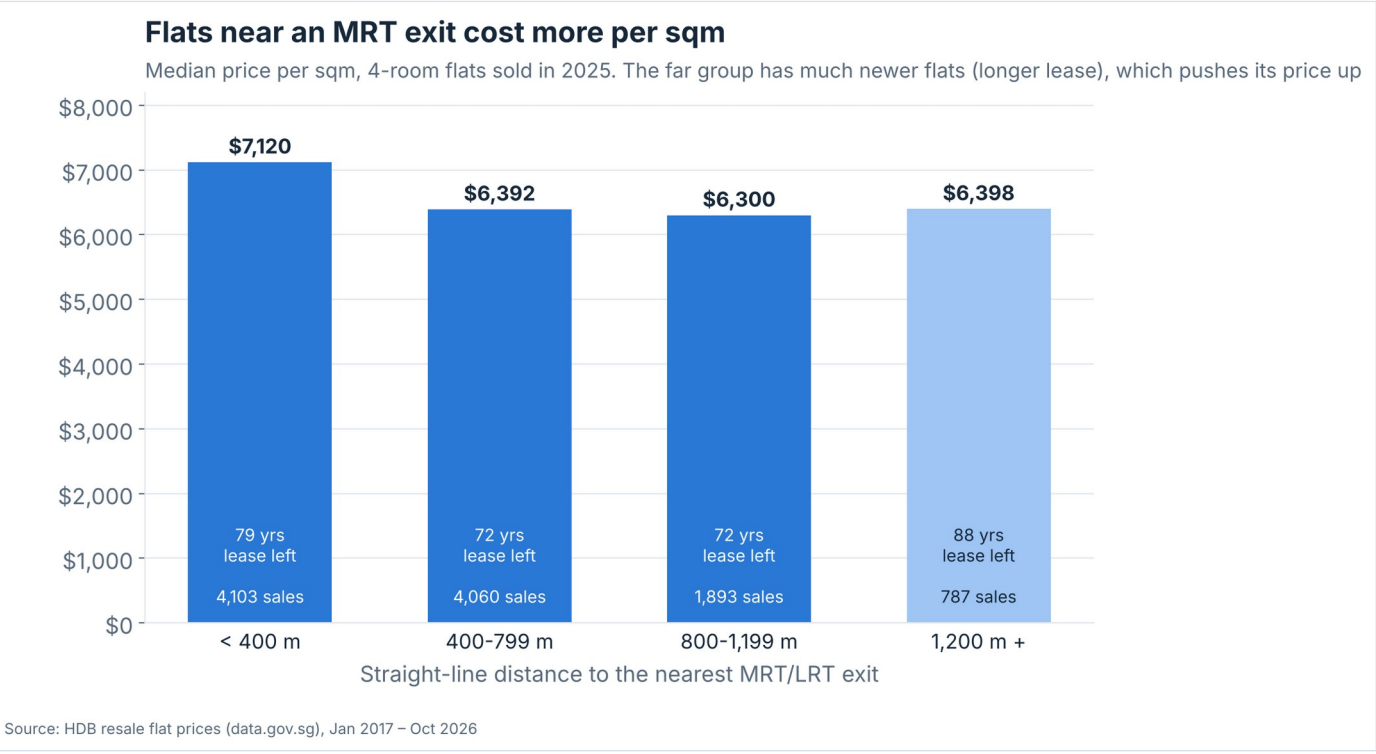
Resale flats sold for S$1 million or more, 2019 and 2025

The whole price range moved up, not only the median.

In 2025, even the cheapest 5% of 4-room flats cost more per sqm (about $5,150) than the 2019 median ($4,188).

Jan to Oct 2026 already has 1,595 such sales.

5 · DRIVERS: DISTANCE TO THE MRT



Source: HDB resale flat prices (data.gov.sg), Jan 2017 – Oct 2026

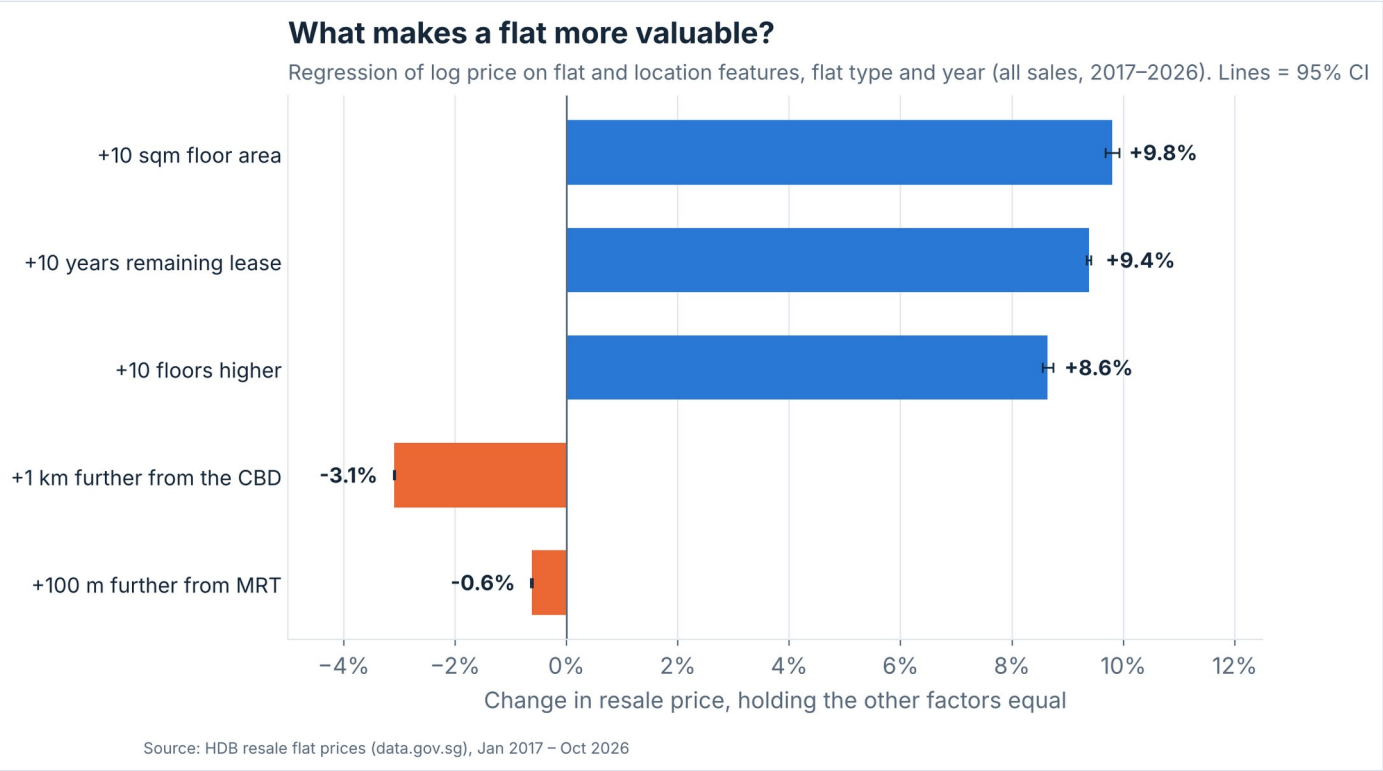
+11%

Price per sqm within 400 m of an MRT exit, compared with 1.2 km or more (4-room, 2025)

One factor alone can mislead: the farthest group is not the cheapest because its flats are newer, with a longer lease.

So we need to hold lease, floor and size equal. The regression does that.

6 · DRIVERS: ALL FACTORS TOGETHER



$R^2 = 0.89$

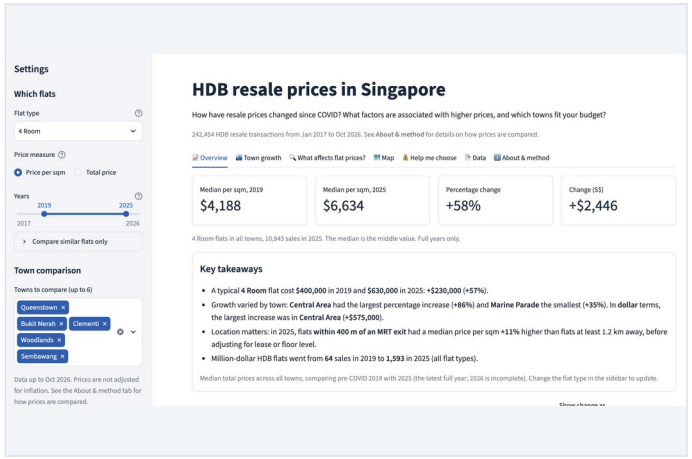
Regression of log price on flat and location features, 242,454 sales, 2017 to 2026

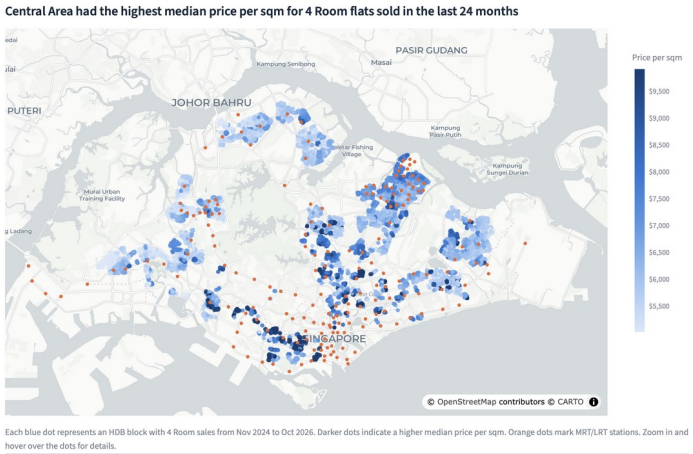
Each step adds about +9%: 10 sqm, 10 years of lease, 10 floors.

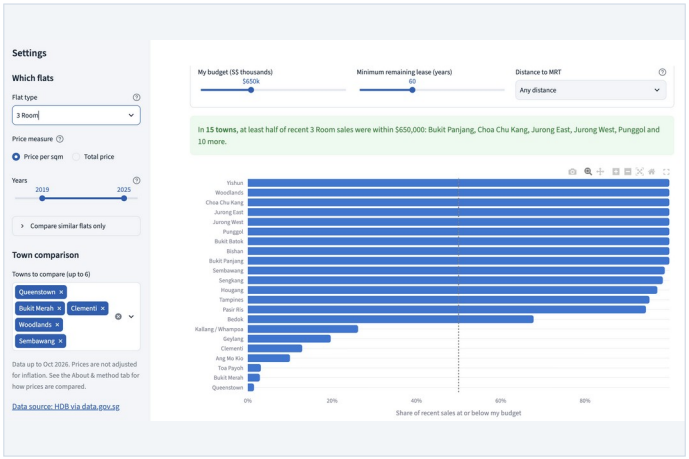
Location still matters: −3.1% per km from the CBD and −0.6% per 100 m from the MRT.

These are associations, not causes. Schools, malls, views and renovation are not in the data.

Overview first, then zoom and filter, then details







Overview

Key takeaways, change since 2019 and the quarterly trend.

Zoom and filter

Filter by flat type, years, towns, lease and floor; the map shows price per sqm by block.

Details on demand

“Help me choose”: set budget, lease and MRT distance to get towns and a block shortlist.

Live dashboard: https://hdb-resale-ds608.streamlit.app

What we changed after feedback from the TA and instructor

Feedback	What we changed
TA: add location drivers	MRT distance, distance to the CBD, map and regression
TA: help users decide	“Help me choose” budget tool
Instructor: use price per sqm	Default measure, compared with 2019 year by year
Instructor: compare like with like	“Compare similar flats only” (remaining lease, floor)
Instructor: 2026 is incomplete	Left out of year-to-year comparisons
Instructor: report % and S$	Both changes shown, with a switch for the ranking
Instructor: explain the method	New “About & method” tab

9 · LIMITATIONS

What the results can and cannot tell us

Sale prices, not rents

The data show what flats sold for, not what they rent for.

Not adjusted for inflation

Part of the +57% is general inflation, so the real rise is smaller.

Mix effects

A town’s median can rise because different flats were sold, such as newer flats entering resale.

Straight-line distances

Walking routes are longer, and today’s MRT map is applied to older sales.

Association, not cause

Schools, malls, views and renovation are not measured.

Only HDB resale flats

New BTO flats and private property are not included.

TAKE-AWAYS

Prices rose sharply, and how you measure growth changes the answer

1. A typical 4-room flat cost 57% more in 2025 than in 2019, and prices have been flat since mid-2025.

2. The towns that “grew most” differ by % and by S$, so the dashboard shows both.

3. Size, lease, floor and location all go with price; distance to the MRT still matters after holding the others equal.

4. With a $650k budget and at least 60 years of lease left, a 3-room buyer still finds 15 towns where at least half of recent sales fit, mostly outside the city centre.

Live dashboard: https://hdb-resale-ds608.streamlit.app
Code, data documentation, AI usage and contributions: github.com/tianjiaoma2026-lab/hdb-resale-dashboard

Thank you. Questions and feedback welcome.

APPENDIX A · METHOD

How we clean, compare and model prices

Cleaning

Month to date; “61 years 04 months” to 61.33 years; “10 TO 12” to floor 11.

Added price per sqm, flat age, mature or non-mature estate (HDB’s list of 15), lease bands and a $1M+ flag.

Comparing years

Medians, same flat type, full years only, each compared with 2019.

% change = (median in year ÷ median in 2019 − 1) × 100. S$ change = median in year − median in 2019.

Locations

Block centres from the NUS 3D HDB dataset (about 90% of blocks), the rest from OneMap. Straight-line distance to the nearest MRT or LRT exit and to Raffles Place.

Regression

log(price) on floor area, remaining lease, floor level, MRT distance, CBD distance, flat type and year. OLS with robust standard errors.

A coefficient b means about 100 × (exp(b) − 1)% change in price.

APPENDIX B · DATA

Data sources

Dataset	Publisher	Used for
HDB resale flat prices, Jan 2017 onwards	HDB via data.gov.sg	All resale transactions
MRT station exits	LTA via data.gov.sg	Location of every MRT/LRT exit
3D city model of HDB buildings	NUS Urban Analytics Lab	Location of about 90% of blocks
OneMap Search API	Singapore Land Authority	Location of newer blocks

Our results use the HDB data downloaded on 12 October 2026. Licences and column descriptions are in data/DATA_SOURCES.md.